

AoPS Pre-Algebra — 90-Question Mastery Test

9 chapters · 10 questions each · moderately hard · answers and explanations at the end

Instructions: aim for 8–12 minutes per chapter. Pick A, B, C, or D. Correct answers are balanced across the four choices.

Chapter 1 — Properties of Arithmetic

1. Simplify using order of operations: $8 - (3 - 7) + 2 \times (-4)$.
A) 4
B) -9
C) -1
D) 13
2. Which property justifies $(a + b) + c = a + (b + c)$?
A) Commutative of addition
B) Associative of addition
C) Distributive
D) Additive identity
3. Use the distributive property to compute $15 \cdot 7 - 15 \cdot 3$.
A) 45
B) 75
C) 60
D) 105
4. Evaluate $-(-(-6)) + 4 \cdot (-2)$.
A) -2
B) 2
C) 10
D) -14
5. Use the distributive property: $12(3 + 5) - 4(3 + 5)$.
A) 64
B) 48
C) 56
D) 96
6. Find $1 + 2 + 3 + \dots + 25$.
A) 300
B) 325
C) 350
D) 375

7. Compute mentally: $999 \cdot 47 + 999 \cdot 3$.
- A) 47,000
 - B) 49,995
 - C) 49,950
 - D) 50,000
8. Which equals 0 by the additive inverse property?
- A) $7 \cdot 0$
 - B) $7 - 0$
 - C) $0/7$
 - D) $7 + (-7)$
9. Evaluate $18 \div 3 \cdot 4 - 2$ (left-to-right after parentheses/exponents).
- A) 22
 - B) 14
 - C) 18
 - D) 26
10. Use the distributive property: $9 \cdot 13 - 9 \cdot 4$.
- A) 63
 - B) 81
 - C) 72
 - D) 117

Chapter 2 — Exponents

11. Simplify $2^3 \cdot 2^5 \div 2^2$.
- A) 2^5
 - B) 2^7
 - C) 2^6
 - D) 2^8
12. Evaluate $(-3)^4$.
- A) -81
 - B) -27
 - C) 27
 - D) 81
13. Compute $5^0 \cdot 3^2 - 2^3$.
- A) 1
 - B) 7
 - C) 9
 - D) 17

14. Write 0.00081 using powers of 3 and 10.

- A) $3^4 \times 10^{-4}$
- B) $3^4 \times 10^{-5}$
- C) $3^3 \times 10^{-4}$
- D) $3^5 \times 10^{-5}$

15. Simplify $(2^2)^3 \cdot 3^0$.

- A) 8
- B) 16
- C) 64
- D) 32

16. Evaluate the negative exponent 4^{-2} .

- A) -16
- B) $1/8$
- C) $-1/8$
- D) $1/16$

17. Compute $2^5 + 2^4 + 2^3$.

- A) 56
- B) 40
- C) 48
- D) 64

18. Which is largest: 3^3 , 4^2 , 5^2 , 2^5 ?

- A) $3^3 = 27$
- B) $2^5 = 32$
- C) $4^2 = 16$
- D) $5^2 = 25$

19. Evaluate $(10^2 - 10) \div 10^1$.

- A) 8
- B) 10
- C) 9
- D) 11

20. Evaluate $(-2)^3 \cdot (-2)^2$.

- A) 32
- B) -16
- C) 16
- D) -32

Chapter 3 — Number Theory

21. Find $\gcd(84, 56)$.

- A) 28
- B) 7
- C) 14
- D) 21

22. Find $\text{lcm}(12, 18)$.

- A) 24
- B) 36
- C) 72
- D) 216

23. Prime factorization of 180.

- A) $2^3 \cdot 3^2 \cdot 5$
- B) $2^2 \cdot 3 \cdot 5^2$
- C) $2^2 \cdot 3^2 \cdot 5$
- D) $2 \cdot 3^2 \cdot 5^2$

24. How many positive divisors does 60 have?

- A) 8
- B) 10
- C) 16
- D) 12

25. Is 91 prime?

- A) No, $91 = 7 \cdot 13$
- B) Yes
- C) No, $91 = 11 \cdot 9$
- D) No, $91 = 3 \cdot 31$

26. Find $\gcd(105, 165)$.

- A) 5
- B) 15
- C) 7
- D) 21

27. How many primes are strictly between 20 and 40?

- A) 3
- B) 5
- C) 4
- D) 6

28. Find $\text{lcm}(8, 20, 30)$.

- A) 60
- B) 180
- C) 240
- D) 120

29. Which prime divides $2^{10} - 1 = 1023$?
- A) 3
 - B) 5
 - C) 7
 - D) 11
30. How many positive integers less than 15 are coprime to 15?
- A) 6
 - B) 8
 - C) 7
 - D) 9

Chapter 4 — Fractions

31. Add $\frac{3}{4} + \frac{5}{6}$.
- A) $\frac{8}{5}$
 - B) $\frac{19}{24}$
 - C) $\frac{19}{12}$
 - D) $1\frac{7}{12}$
32. Multiply $2\frac{1}{3} \times \frac{3}{7}$.
- A) $\frac{10}{7}$
 - B) $\frac{7}{3}$
 - C) $1\frac{1}{7}$
 - D) 1
33. Compute $\frac{5}{8} \div \frac{3}{4}$.
- A) $\frac{5}{6}$
 - B) $\frac{15}{32}$
 - C) $\frac{2}{3}$
 - D) $\frac{3}{5}$
34. Simplify $(1 + \frac{1}{2}) \div (1 - \frac{1}{3})$.
- A) $\frac{5}{4}$
 - B) $\frac{9}{4}$
 - C) $\frac{3}{2}$
 - D) 2
35. Subtract $\frac{7}{9} - \frac{4}{15}$.
- A) $\frac{11}{45}$
 - B) $\frac{19}{45}$
 - C) $\frac{23}{45}$
 - D) $\frac{31}{45}$

36. Compute $4\frac{2}{5} - 1\frac{7}{8}$.

- A) $2\frac{11}{40}$
- B) $2\frac{19}{40}$
- C) $3\frac{3}{40}$
- D) $2\frac{21}{40}$

37. Reciprocal of $3\frac{1}{4}$.

- A) $\frac{4}{13}$
- B) $\frac{13}{4}$
- C) $\frac{3}{13}$
- D) $\frac{13}{3}$

38. Which fraction is the smallest: $\frac{5}{7}$, $\frac{7}{9}$, $\frac{8}{11}$, $\frac{4}{5}$?

- A) $\frac{7}{9}$
- B) $\frac{5}{7}$
- C) $\frac{8}{11}$
- D) $\frac{4}{5}$

39. Compute $(\frac{2}{3})$ of $(\frac{5}{4})$ of 36.

- A) 20
- B) 45
- C) 30
- D) 60

40. Solve $\frac{x}{5} + \frac{1}{2} = \frac{7}{10}$.

- A) $\frac{1}{2}$
- B) $\frac{3}{2}$
- C) 2
- D) 1

Chapter 5 — Equations & Inequalities

41. Solve $3x - 7 = 14$.

- A) $x = 7$
- B) $x = 5$
- C) $x = 6$
- D) $x = \frac{21}{3}$

42. Solve $2(x + 4) = 5x - 3$.

- A) $x = 5$
- B) $x = \frac{11}{3}$
- C) $x = 7$
- D) $x = 11$

43. Solve the inequality $4x + 2 > 18$.

- A) $x \geq 4$
- B) $x > 3$
- C) $x > 4$
- D) $x < 4$

44. Solve $|2x - 1| = 7$.

- A) $x = 3$ or $x = -4$
- B) $x = 4$ only
- C) $x = -3$ only
- D) $x = 4$ or $x = -3$

45. Twice a number minus 5 equals 11. Find the number.

- A) 8
- B) 6
- C) 7
- D) 9

46. Solve $5 - 3(x + 1) \leq 2$.

- A) $x \leq 0$
- B) $x \geq 0$
- C) $x \leq 2/3$
- D) $x \geq 2/3$

47. Solve the system $x + y = 10$, $x - y = 4$.

- A) (5, 5)
- B) (6, 4)
- C) (7, 3)
- D) (8, 2)

48. Solve $(1/2)x + 3/4 = 5/4$.

- A) $x = 1/2$
- B) $x = 2$
- C) $x = 3$
- D) $x = 1$

49. Solve the compound inequality $2 < 3x - 1 < 8$.

- A) $1 < x < 3$
- B) $0 < x < 3$
- C) $1 < x \leq 3$
- D) $1 \leq x \leq 3$

50. Solve for y: $2y = 8 - 3(y + 1)$.

- A) $y = -1$
- B) $y = 1$
- C) $y = 2$
- D) $y = 5/2$

Chapter 6 — Decimals

51. Compute $0.75 + 1.2 - 0.375$.

- A) 1.2
- B) 1.95
- C) 1.575
- D) 2.325

52. Compute $0.8 \times 0.25 \times 4$.

- A) 0.2
- B) 0.5
- C) 1.0
- D) 0.8

53. Compute $4.8 \div 0.6$.

- A) 8
- B) 0.8
- C) 80
- D) 0.08

54. Express $0.454545\dots$ (the repeating decimal $0.\overline{45}$) as a fraction.

- A) $\frac{9}{20}$
- B) $\frac{5}{11}$
- C) $\frac{45}{100}$
- D) $\frac{5}{9}$

55. Compute 12% of 250.

- A) 18
- B) 25
- C) 30
- D) 32

56. Round 3.14159 to the hundredths place.

- A) 3.142
- B) 3.15
- C) 3.141
- D) 3.14

57. Compute $0.6 \times (1.5 + 2.5)$.

- A) 2.4
- B) 1.8
- C) 3.0
- D) 4.2

58. Which is largest: 0.099, 0.101, 0.1, 0.098?

- A) 0.098
- B) 0.101
- C) 0.099
- D) 0.1

59. Compute $5.25 \div 0.75$.

- A) 6
- B) 6.5
- C) 7
- D) 7.5

60. In scientific notation: $4.5 \times 10^3 + 2.5 \times 10^2$.

- A) 4.7×10^3
- B) 4.75×10^2
- C) 4.75×10^4
- D) 4.75×10^3

Chapter 7 — Ratios, Rates & Conversions

61. Split 40 in the ratio 3 : 5. The two parts are:

- A) 15 : 25
- B) 12 : 28
- C) 12 : 20
- D) 18 : 22

62. A car covers 240 km in 4 hours. Average speed?

- A) 40 km/h
- B) 60 km/h
- C) 50 km/h
- D) 80 km/h

63. Convert 2.5 L to mL.

- A) 25
- B) 250
- C) 2,500
- D) 25,000

64. A map uses 1 cm : 5 km. What real distance does 12 cm represent?

- A) 40 km
- B) 55 km
- C) 65 km
- D) 60 km

65. Solve the proportion $\frac{3}{8} = \frac{x}{48}$.
- A) 18
 - B) 12
 - C) 16
 - D) 24
66. A box of 6 items costs \$18. Unit price?
- A) \$2
 - B) \$3
 - C) \$2.50
 - D) \$3.50
67. Increase 200 by 15%.
- A) 215
 - B) 220
 - C) 230
 - D) 225
68. If 5 apples cost \$2, how many apples cost \$10?
- A) 15
 - B) 20
 - C) 30
 - D) 25
69. Convert 45 mph to km/h (use $1 \text{ mi} \approx 1.6 \text{ km}$).
- A) 72
 - B) 64
 - C) 70
 - D) 80
70. $a : b : c = 2 : 3 : 5$, and $a + b + c = 50$. Find the largest part.
- A) 10
 - B) 25
 - C) 15
 - D) 20

Chapter 9 — Percents & Square Roots

71. Compute 35% of 420.
- A) 120
 - B) 135
 - C) 147
 - D) 150

72. 24 is what percent of 80?

- A) 20%
- B) 25%
- C) 35%
- D) 30%

73. Compute $\sqrt{64} + \sqrt{81}$.

- A) 17
- B) 15
- C) 23
- D) 145

74. Decrease 120 by 25%.

- A) 80
- B) 90
- C) 95
- D) 100

75. Simplify $\sqrt{(36/49)}$.

- A) 7/6
- B) 18/7
- C) 6/7
- D) 36/7

76. Simple interest on \$500 at 4% for 2 years.

- A) \$20
- B) \$42
- C) \$80
- D) \$40

77. Compute $\sqrt{121} \times \sqrt{4}$.

- A) 22
- B) 11
- C) 44
- D) 242

78. 80 is what percent of 200?

- A) 25%
- B) 40%
- C) 30%
- D) 50%

79. Approximate $\sqrt{50}$.

- A) 6.93
- B) 7.00
- C) 7.07
- D) 7.20

80. Percent change from 50 to 65.

- A) 15%
- B) 25%
- C) 35%
- D) 30%

Chapter 14 — Statistics, Data & Probability

81. Mean of 12, 15, 18, 9, 21.

- A) 15
- B) 14
- C) 16
- D) 17

82. Median of 3, 7, 8, 12, 2, 10.

- A) 7
- B) 7.5
- C) 8
- D) 9

83. Mode of 4, 4, 5, 6, 4, 7.

- A) 5
- B) 6
- C) 4
- D) No mode

84. Range of 45, 52, 38, 61.

- A) 16
- B) 17
- C) 29
- D) 23

85. Probability of rolling an even number on a fair 6-sided die.

- A) $\frac{1}{2}$
- B) $\frac{1}{6}$
- C) $\frac{1}{3}$
- D) $\frac{2}{3}$

86. Five bars on a graph have heights summing to 120. Mean height?

- A) 20
- B) 24
- C) 25
- D) 30

87. Which description best fits a data set with low standard deviation?
- A) Many modes
 - B) Large spread
 - C) Values close to the mean
 - D) High range
88. Where is the lower quartile (25%) in an ordered list of 20 values (using rank $n \cdot (p)/100$ rounded up)?
- A) 4th value
 - B) 10th value
 - C) 15th value
 - D) 5th–6th value
89. Group A: 10 students, mean 80. Group B: 10 students, mean 90. Combined mean?
- A) 85
 - B) 82
 - C) 87
 - D) 88
90. A spinner lands on red with probability $1/4$. Probability of red on all three independent spins?
- A) $1/12$
 - B) $1/64$
 - C) $1/16$
 - D) $3/4$

Answer Key & Explanations

Answer distribution — A: 23, B: 23, C: 22, D: 22 (total 90)

Chapter 1 — Properties of Arithmetic

1. Correct answer: **A**. Inside parens: $3 - 7 = -4$. So $8 - (-4) + 2(-4) = 8 + 4 - 8 = 4$.
2. Correct answer: **B**. Associative property regroups terms without changing their order.
3. Correct answer: **C**. $15(7 - 3) = 15 \cdot 4 = 60$.
4. Correct answer: **D**. $-(-(-6)) = -6$ and $4(-2) = -8$, so $-6 + (-8) = -14$.
5. Correct answer: **A**. $(12 - 4)(3 + 5) = 8 \cdot 8 = 64$.
6. Correct answer: **B**. Sum = $n(n+1)/2 = 25 \cdot 26/2 = 325$.
7. Correct answer: **C**. $999(47 + 3) = 999 \cdot 50 = 49,950$.
8. Correct answer: **D**. Additive inverse: $a + (-a) = 0$. Choice B uses the zero product rule.
9. Correct answer: **A**. $18/3 = 6$, then $6 \cdot 4 = 24$, then $24 - 2 = 22$.
10. Correct answer: **B**. $9(13 - 4) = 9 \cdot 9 = 81$.

Chapter 2 — Exponents

11. Correct answer: **C**. Add exponents in product, subtract for quotient: $3 + 5 - 2 = 6$.
12. Correct answer: **D**. Even exponent $\Rightarrow$ positive: $3 \cdot 3 \cdot 3 \cdot 3 = 81$.
13. Correct answer: **A**. $5^0 = 1$, so $1 \cdot 9 - 8 = 1$.
14. Correct answer: **B**. $0.00081 = 81/100000 = 3^4/10^5 = 3^4 \times 10^{-5}$.
15. Correct answer: **C**. $(2^2)^3 = 2^6 = 64$; $3^0 = 1$; product 64.
16. Correct answer: **D**. $a^{-1} = 1/a$, so $4^{-2} = 1/16$.
17. Correct answer: **A**. $32 + 16 + 8 = 56$.
18. Correct answer: **B**. $27, 16, 25, 32 \Rightarrow 2^5$ wins.
19. Correct answer: **C**. $(100 - 10)/10 = 90/10 = 9$.
20. Correct answer: **D**. $(-2)^5 = -32$ because the exponent is odd.

Chapter 3 — Number Theory

21. Correct answer: **A**. $84 = 2^2 \cdot 3 \cdot 7$, $56 = 2^3 \cdot 7 \Rightarrow \text{gcd} = 2^2 \cdot 7 = 28$.
22. Correct answer: **B**. $12 = 2^2 \cdot 3$, $18 = 2 \cdot 3^2 \Rightarrow \text{lcm} = 2^2 \cdot 3^2 = 36$.
23. Correct answer: **C**. $180 = 4 \cdot 45 = 4 \cdot 9 \cdot 5 = 2^2 \cdot 3^2 \cdot 5$.
24. Correct answer: **D**. $60 = 2^2 \cdot 3 \cdot 5$; divisor count = $3 \cdot 2 \cdot 2 = 12$.
25. Correct answer: **A**. $91 = 7 \cdot 13$, so it is composite.
26. Correct answer: **B**. $105 = 3 \cdot 5 \cdot 7$, $165 = 3 \cdot 5 \cdot 11 \Rightarrow \text{gcd} = 15$.
27. Correct answer: **C**. $23, 29, 31, 37 \Rightarrow 4$ primes.
28. Correct answer: **D**. $8 = 2^3$, $20 = 2^2 \cdot 5$, $30 = 2 \cdot 3 \cdot 5 \Rightarrow \text{lcm} = 2^3 \cdot 3 \cdot 5 = 120$.

29. Correct answer: **A.** $1023 = 3 \cdot 11 \cdot 31$, so 3 divides it (so does 11, but 3 is smallest listed).

30. Correct answer: **B.** $\phi(15) = 15(1 - 1/3)(1 - 1/5) = 8$ (namely 1,2,4,7,8,11,13,14).

Chapter 4 — Fractions

31. Correct answer: **C.** LCD 12: $9/12 + 10/12 = 19/12$ (same as $1 \frac{7}{12}$, but pick the reduced improper form).

32. Correct answer: **D.** $(7/3)(3/7) = 21/21 = 1$.

33. Correct answer: **A.** $5/8 \cdot 4/3 = 20/24 = 5/6$.

34. Correct answer: **B.** $(3/2)/(2/3) = (3/2)(3/2) = 9/4$.

35. Correct answer: **C.** LCD 45: $35/45 - 12/45 = 23/45$.

36. Correct answer: **D.** $22/5 - 15/8 = 176/40 - 75/40 = 101/40 = 2 \frac{21}{40}$.

37. Correct answer: **A.** $3 \frac{1}{4} = 13/4$, so the reciprocal is $4/13$.

38. Correct answer: **B.** Decimals: 0.714, 0.778, 0.727, 0.800 $\Rightarrow$ 5/7 smallest.

39. Correct answer: **C.** $(2/3)(5/4)(36) = (10/12)(36) = 30$.

40. Correct answer: **D.** $x/5 = 7/10 - 5/10 = 2/10 = 1/5$, so $x = 1$.

Chapter 5 — Equations & Inequalities

41. Correct answer: **A.** $3x = 21$, so $x = 7$.

42. Correct answer: **B.** $2x + 8 = 5x - 3 \Rightarrow 3x = 11 \Rightarrow x = 11/3$.

43. Correct answer: **C.** $4x > 16 \Rightarrow x > 4$ (strict).

44. Correct answer: **D.** $2x - 1 = \pm 7$ gives $x = 4$ or $x = -3$.

45. Correct answer: **A.** $2n - 5 = 11 \Rightarrow 2n = 16 \Rightarrow n = 8$.

46. Correct answer: **B.** $5 - 3x - 3 \leq 2 \Rightarrow 2 - 3x \leq 2 \Rightarrow -3x \leq 0 \Rightarrow x \geq 0$.

47. Correct answer: **C.** Add: $2x = 14$, $x = 7$, $y = 3$.

48. Correct answer: **D.** $(1/2)x = 1/2$, so $x = 1$.

49. Correct answer: **A.** Add 1: $3 < 3x < 9$, divide: $1 < x < 3$.

50. Correct answer: **B.** $2y = 8 - 3y - 3 = 5 - 3y \Rightarrow 5y = 5 \Rightarrow y = 1$.

Chapter 6 — Decimals

51. Correct answer: **C.** $0.75 + 1.2 = 1.95$, then $1.95 - 0.375 = 1.575$.

52. Correct answer: **D.** $0.8 \times 0.25 = 0.2$, then $0.2 \times 4 = 0.8$.

53. Correct answer: **A.** Multiply both by 10: $48 \div 6 = 8$.

54. Correct answer: **B.** Let $x = 0.4545\dots$; $100x - x = 45$, so $99x = 45$, $x = 45/99 = 5/11$.

55. Correct answer: **C.** $0.12 \times 250 = 30$.

56. Correct answer: **D.** Hundredths digit is 4; the next digit is 1 (< 5), so 3.14.

57. Correct answer: **A.** $1.5 + 2.5 = 4$, then $0.6 \times 4 = 2.4$.

58. Correct answer: **B.** $0.101 > 0.100 > 0.099 > 0.098$.

59. Correct answer: **C**. $525 \div 75 = 7$.

60. Correct answer: **D**. $4500 + 250 = 4750 = 4.75 \times 10^3$.

Chapter 7 — Ratios, Rates & Conversions

61. Correct answer: **A**. Each share is $40/8 = 5$; parts $3 \cdot 5 = 15$ and $5 \cdot 5 = 25$.

62. Correct answer: **B**. $240/4 = 60$ km/h.

63. Correct answer: **C**. 1 L = 1000 mL, so $2.5 \times 1000 = 2,500$ mL.

64. Correct answer: **D**. $12 \times 5 = 60$ km.

65. Correct answer: **A**. $x = 48 \times 3/8 = 18$.

66. Correct answer: **B**. $18/6 = \$3$ each.

67. Correct answer: **C**. $200 \times 1.15 = 230$.

68. Correct answer: **D**. $\$10/\$2 = 5$ groups of 5 apples = 25.

69. Correct answer: **A**. $45 \times 1.6 = 72$ km/h.

70. Correct answer: **B**. Each share is $50/10 = 5$, so largest = $5 \cdot 5 = 25$.

Chapter 9 — Percents & Square Roots

71. Correct answer: **C**. $0.35 \times 420 = 147$.

72. Correct answer: **D**. $24/80 = 0.30 = 30\%$.

73. Correct answer: **A**. $8 + 9 = 17$.

74. Correct answer: **B**. $120 - 30 = 90$.

75. Correct answer: **C**. $\sqrt{36}/\sqrt{49} = 6/7$.

76. Correct answer: **D**. $I = Prt = 500 \cdot 0.04 \cdot 2 = \40 .

77. Correct answer: **A**. $11 \times 2 = 22$.

78. Correct answer: **B**. $80/200 = 0.40 = 40\%$.

79. Correct answer: **C**. $\sqrt{50} = 5\sqrt{2} \approx 5 \cdot 1.4142 \approx 7.07$.

80. Correct answer: **D**. $(65 - 50)/50 = 15/50 = 30\%$.

Chapter 14 — Statistics, Data & Probability

81. Correct answer: **A**. Sum 75, divided by 5 = 15.

82. Correct answer: **B**. Sorted: 2, 3, 7, 8, 10, 12. Middle pair (7, 8) $\Rightarrow$ median 7.5.

83. Correct answer: **C**. 4 appears three times — most frequent value.

84. Correct answer: **D**. Max – min = $61 - 38 = 23$.

85. Correct answer: **A**. Even outcomes {2, 4, 6}; $3/6 = 1/2$.

86. Correct answer: **B**. $120/5 = 24$.

87. Correct answer: **C**. Low standard deviation means small dispersion around the mean.

88. Correct answer: **D**. Position $20 \cdot 0.25 = 5$, so Q1 is between the 5th and 6th values.

89. Correct answer: **A.** Total sum = $800 + 900 = 1700$, over 20 $\Rightarrow 85$.

90. Correct answer: **B.** $(1/4)^3 = 1/64$.